

Chapter 1.

INTRODUCTION TO DEEP LEARNING

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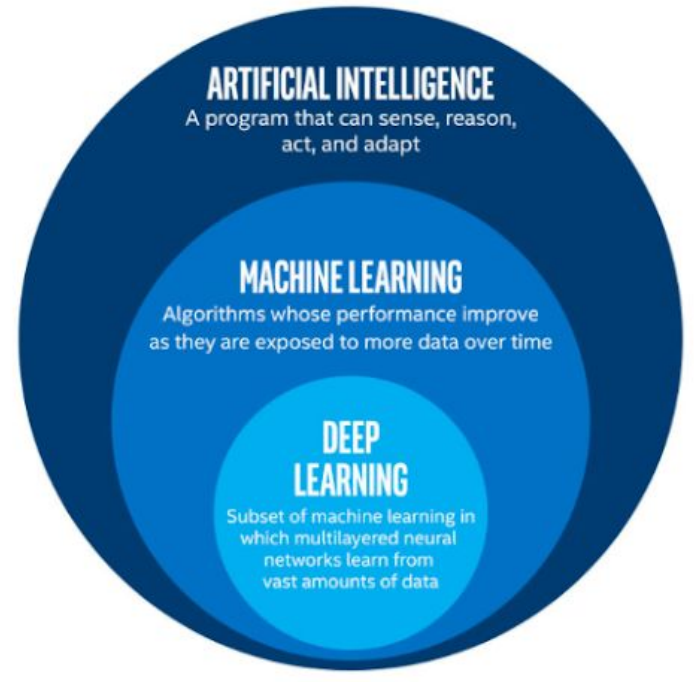


Key concepts of Deep learning

Deep learning is a subfield of machine learning that has revolutionized artificial intelligence.

Core Concepts:

- **Neural Networks:**
 - At the heart of deep learning are artificial neural networks, inspired by the structure of the human brain.
 - These networks consist of interconnected nodes (neurons) organized in layers:
 - **Input Layer:** Receives the raw data.
 - **Hidden Layers:** Perform complex computations on the data. Deep learning networks have multiple hidden layers, hence the "deep" in deep learning.
 - **Output Layer:** Produces the final result.
- **Deep Neural Networks:**
 - The "deep" in deep learning refers to the multiple layers in these neural networks.
 - These layers enable the network to learn increasingly complex representations of data.



INTRODUCTION TO DEEP LEARNING

Deep Learning is a subset of machine learning in artificial intelligence that uses layered neural networks to analyze various factors of data. It automates predictive analytics by simulating the workings of the human brain.

APPLICATIONS OF DEEP LEARNING



Financial Services



Healthcare



Customer Service

CORE COMPONENTS OF DEEP LEARNING



Neural Networks

Neural networks are the core entities that power deep learning models. They are actually computational models made of interconnected nodes, similar to neurons of the brain.



Deep Neural Networks

Deep neural networks are neural networks that have multiple layers, which allow them to learn the more complex features and arrive at accurate predictions.



Activation Functions

Activation functions are responsible for deciding which information must be passed to the next layer. They introduce complexity & help the network learn from data.

HOW DEEP LEARNING WORKS



Input



Features Extraction + Classification

The input image is a cartoon representation of a Labrador Retriever.

Output

BENEFITS AND CHALLENGES

Accuracy: Improves as more data is provided.

Automation: Can automate complex decision-making tasks.

Adaptability: Can learn and adapt to new data without human intervention.



Data Requirements: Requires massive amounts of data to train effectively.

Computational Cost: Needs significant computing power and resources.

Black Box Nature: Often, how decisions are made is not transparent.

Categories of Python Libraries for AI/ML Development

Data Manipulation and Analysis



NumPy



Pandas

Visualization



Matplotlib



Seaborn

Machine Learning



Scikit-learn



XGBoost

Natural Language Processing



Gensim



Hugging face

Deep Learning



Tensorflow



PyTorch

Computer Vision



OpenCV

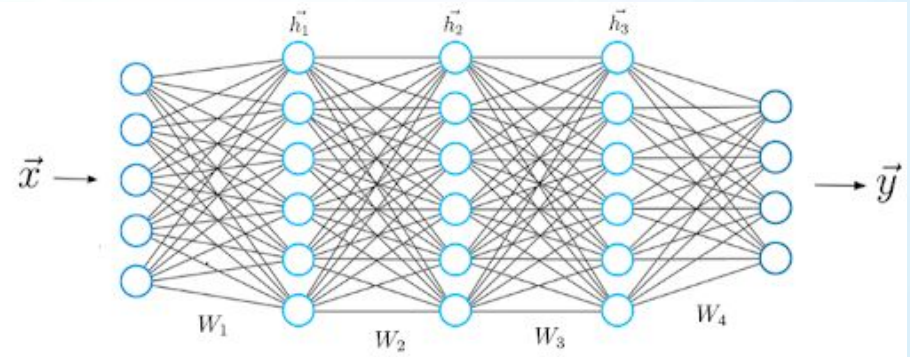
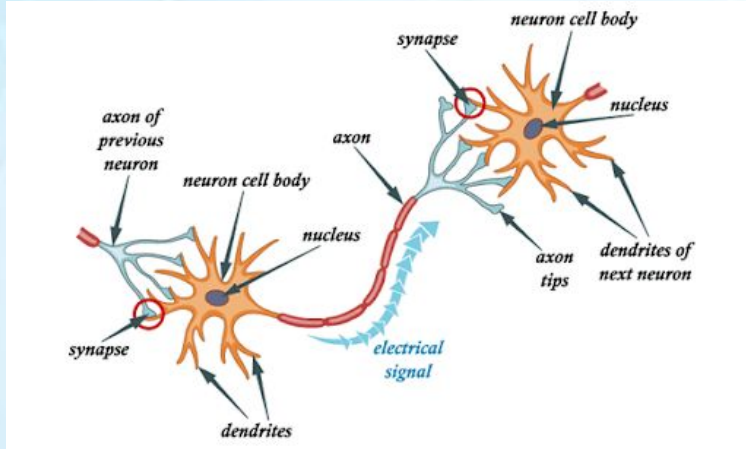


DLib

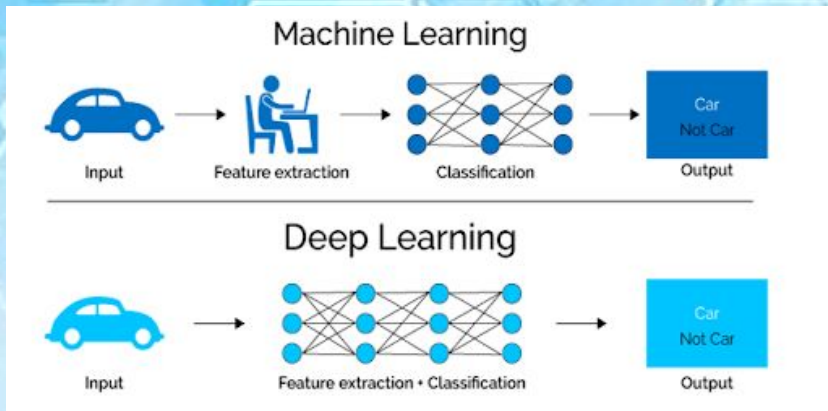
Key concepts of Deep learning

How it Works:

- Deep learning models learn by processing vast amounts of data.
- They automatically extract features from the data, eliminating the need for manual feature engineering.
- The network adjusts the connections (weights) between neurons through a process called backpropagation, minimizing the difference between predicted and actual outputs.



Key Differences from Machine Learning

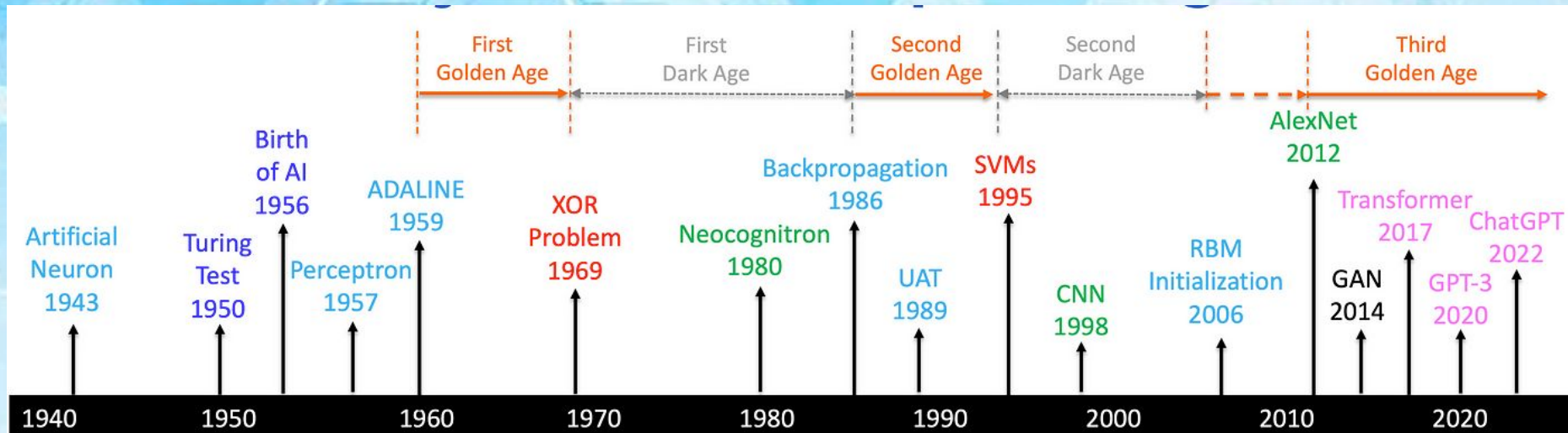


Key Differences from Machine Learning:

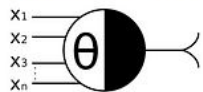
- **Feature Extraction:** Traditional machine learning often requires manual feature extraction, while deep learning automates this process.
- **Data Requirements:** Deep learning models typically require large amounts of data to perform effectively.
- **Complexity:** Deep learning models can handle more complex problems and data than many traditional machine learning algorithms.

	Deep Learning	Machine Learning
Data	Needs a big dataset	Performs well with a small to a medium dataset
Hardware requirements	Requires machines with GPU	Works with low-end machines
Engineering peculiarities	Needs to understand the basic functionality of the data	Understands the features and how they represent the data
Training time	Long	Short
Processing time	A few hours or weeks	A few seconds or hours
Number of algorithms	Few	Many
Data interpretation	Difficult	Some ML algorithms are easy to interpret, whereas some are hardly possible

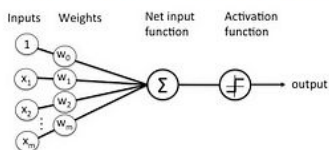
History of Deep Learning



McCulloch-Pitts

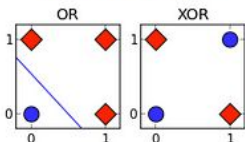


Rosenblatt

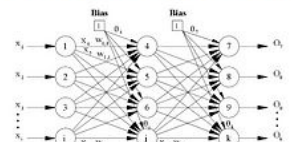


Widrow-Hoff

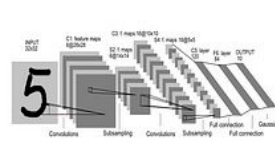
Minsky-Papert



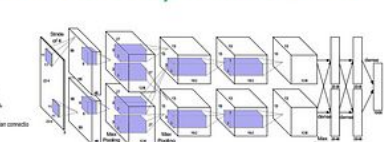
Rumelhart, Hinton et al.



LeCun



Hinton-Ruslan



Krizhevsky et al.

Vaswani

History of Deep Learning

- **Early Foundations (1940s-1960s):**

- The seeds were sown with the work of Walter Pitts and Warren McCulloch, who proposed a computational model of neural networks in 1943.
- Frank Rosenblatt's Perceptron (1958) introduced the concept of trainable artificial neurons.
- Early challenges, like the limitations of the Perceptron and the "AI winter," hindered progress.

- **Resurgence (1980s-1990s):**

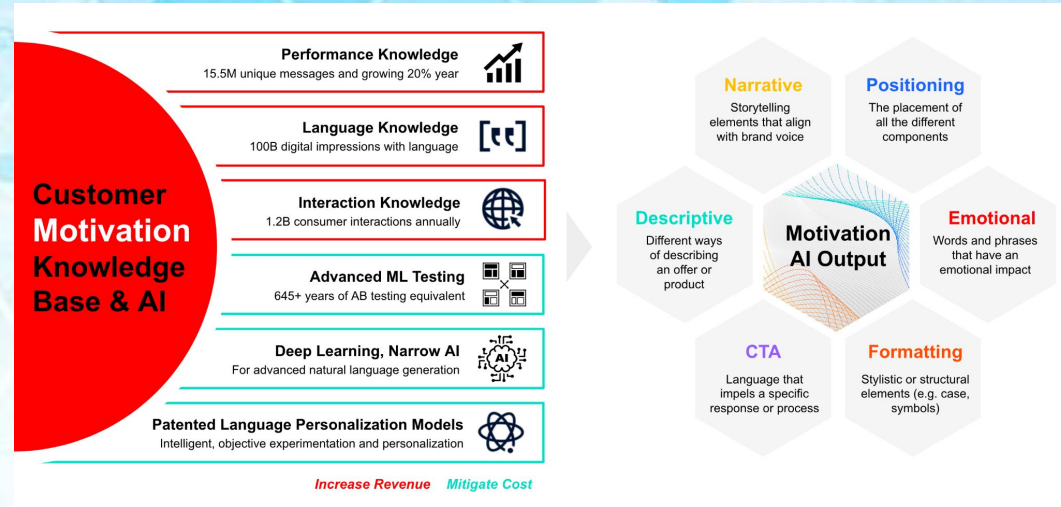
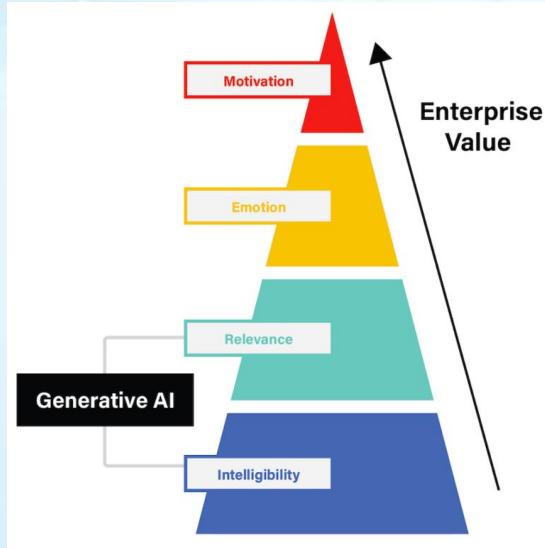
- The development of the backpropagation algorithm in the 1980s was a crucial breakthrough, enabling the training of multi-layer neural networks.
- Yann LeCun's work on Convolutional Neural Networks (CNNs) for handwritten digit recognition (LeNet) demonstrated the practical potential of deep learning.

- **The Deep Learning Revolution (2000s-Present):**

- Increased computational power, particularly the use of GPUs, made it possible to train much larger and deeper networks.
- Datasets like ImageNet provided the vast amounts of data needed for effective training.
- Breakthroughs in areas like computer vision (AlexNet), natural language processing (NLP), and reinforcement learning (AlphaGo) propelled deep learning into the mainstream.
- The creation of transformer models have revolutionized NLP, and many other fields.

Motivation of Deep Learning

- **Mimicking the Brain.** Deep learning is inspired by the hierarchical structure of the human brain, aiming to create systems that can learn complex patterns and representations from data.
- **Automating Feature Extraction.** Traditional machine learning often requires manual feature engineering, which can be time-consuming and challenging. Deep learning automates this process, allowing models to learn relevant features directly from raw data.



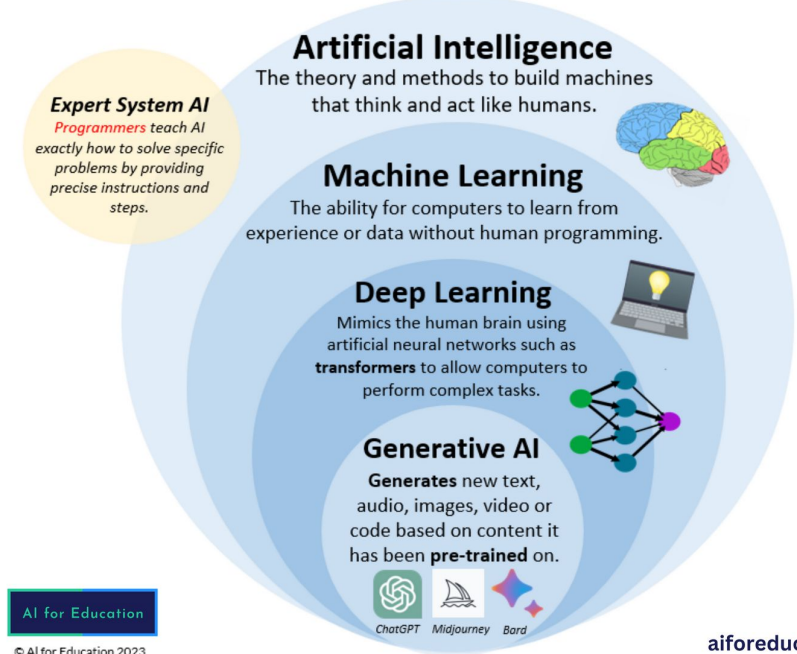
- **Solving Complex Problems.** Deep learning has proven highly effective in tackling problems that are difficult for traditional algorithms, such as image recognition, speech recognition, and natural language understanding.
- **The need to process and understand vast amounts of data.** The explosion of digital information has created a need for algorithms that can make sense of that data. Deep learning excels at this.

Evolution of Deep Learning

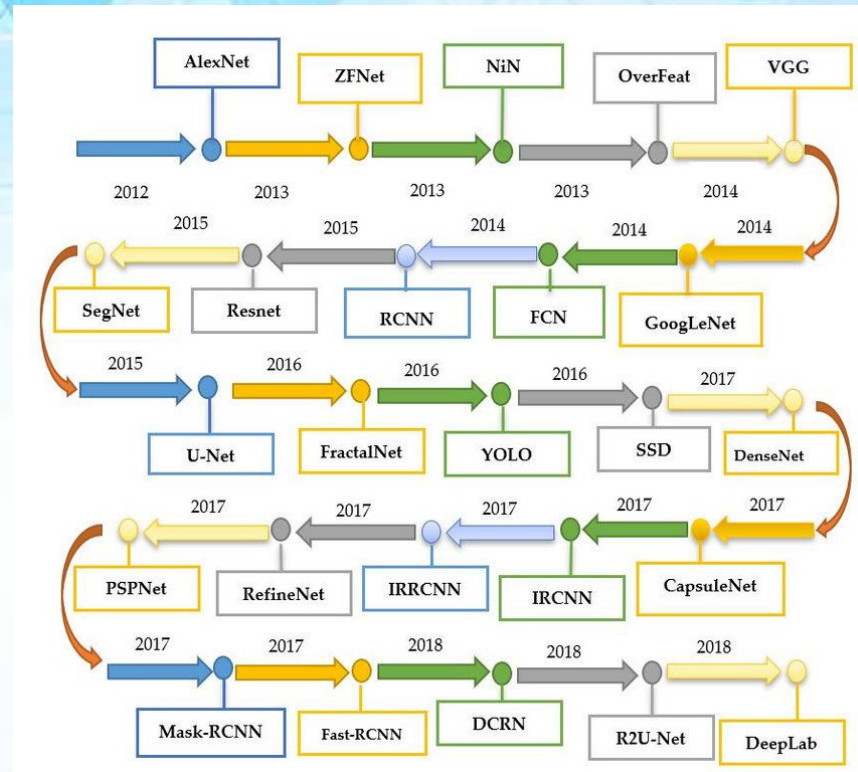
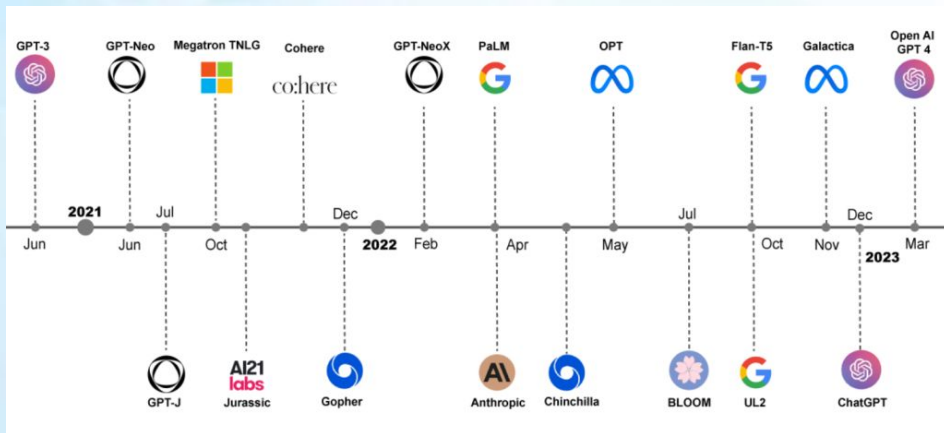
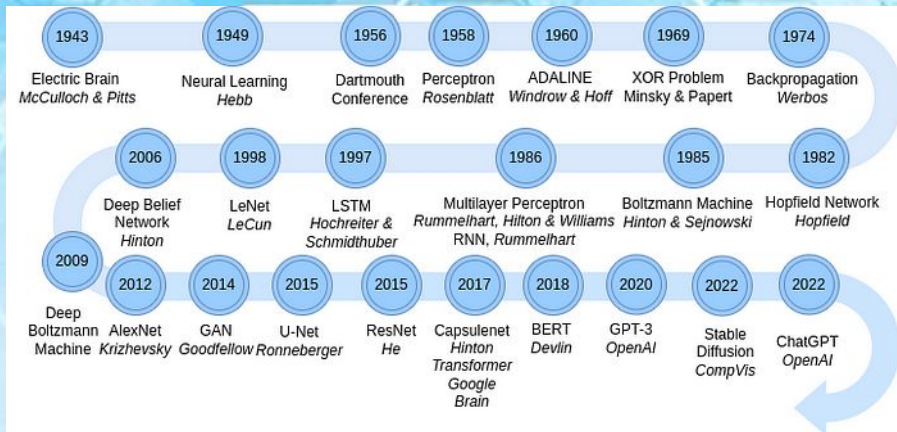
- **Architectural Innovations.** The development of various neural network architectures, such as CNNs, Recurrent Neural Networks (RNNs), and Transformers, has enabled deep learning to address a wide range of tasks.
- **Algorithmic Advancements.** Ongoing research has led to improvements in training algorithms, optimization techniques, and regularization methods.
- **Hardware Acceleration.** The development of specialized hardware, such as GPUs and TPUs, has significantly accelerated deep learning training and inference.
- **Data Availability.** The increase in the amount of available data has been a key factor in the success of deep learning.
- **Expanding Applications.** Deep learning is now being applied in diverse fields, including healthcare, finance, transportation, and entertainment.
- **Ethical Considerations.** As deep learning becomes more powerful, there is an increasing focus on addressing ethical concerns, such as bias, fairness, and privacy.

Defining Generative AI

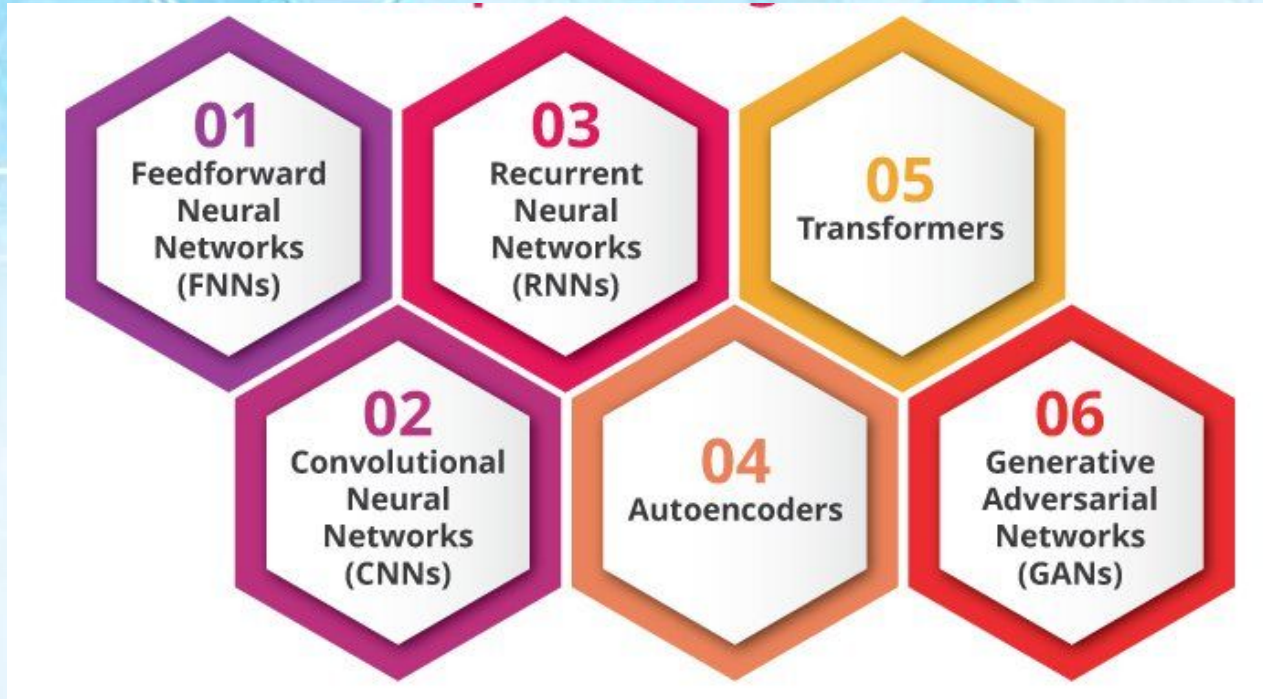
To understand generative artificial intelligence (GenAI), we first need to understand how the technology builds from each of the AI subcategories listed below.



Evolution of Deep Learning



Types of DL Networks



Types of DL Networks

The most common types:

1. Convolutional Neural Networks (CNNs):

- **Best for:** Image recognition, image classification, object detection, video analysis, and other visual tasks.
- **Key features:** Convolutional layers (for feature extraction), pooling layers (for dimensionality reduction), and fully connected layers (for classification).
- **Examples:** ResNet, VGGNet, AlexNet, Efficient Net.

2. Recurrent Neural Networks (RNNs):

- **Best for:** Processing sequential data, such as natural language processing (NLP), time series analysis, and speech recognition.
- **Key features:** Recurrent connections that allow information to persist over time.
- **Examples:** LSTM (Long Short-Term Memory), GRU (Gated Recurrent Unit).

3. Transformers:

- **Best for:** NLP tasks (like machine translation, text summarization, and question answering), but also increasingly used in computer vision.
 - **Key features:** Attention mechanisms that allow the model to focus on relevant parts of the input sequence.
- Examples:** BERT, GPT, Transformer, Vision Transformer (ViT).

4. Generative Adversarial Networks (GANs):

- **Best for:** Generating new data, such as images, music, and text.
- **Key features:** Two networks (a generator and a discriminator) that compete against each other.
- **Examples:** DCGAN, StyleGAN.

5. Autoencoders:

- **Best for:** Dimensionality reduction, feature learning, and anomaly detection.
- **Key features:** An encoder that compresses the input data and a decoder that reconstructs it.

6. Deep Belief Networks (DBNs):

- **Best for:** Unsupervised learning and feature extraction.
- **Key features:** A stack of Restricted Boltzmann Machines (RBMs).

7. Graph Neural Networks (GNNs):

- **Best for:** Working with graph-structured data, such as social networks, molecular structures, and knowledge graphs.
- **Key features:** Message passing between nodes in a graph.

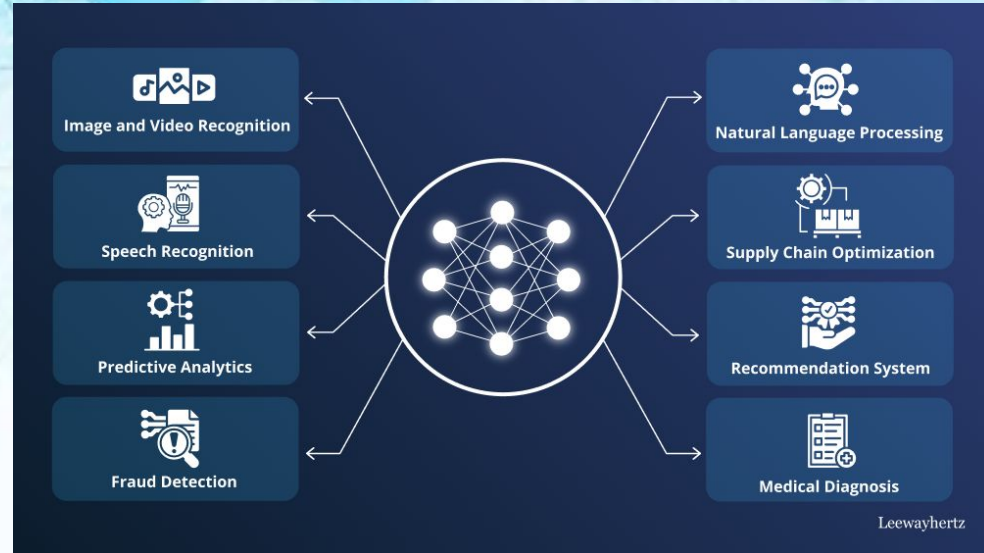
Application of Deep learning

Applications. Deep learning has led to significant advances in various fields, including:

- **Image Recognition:** Identifying objects, people, and scenes in images and videos.
- **Natural Language Processing (NLP):** Understanding and generating human language, powering applications like chatbots and machine translation.
- **Speech Recognition:** Converting spoken language into text.
- **Autonomous Driving:** Enabling vehicles to perceive their surroundings and navigate safely.
- **Medical Diagnosis:** Assisting in the detection of diseases from medical images.

Key Considerations:

- **Data Availability**
- **Computational Resources**
- **"Black Box" Nature:** Deep learning models can be difficult to interpret, making it challenging to understand how they arrive at their predictions.



The main problems addressed by deep learning

Deep learning tackles a wide range of complex problems, often categorized as follows:

1. Classification:

- **Description:**

- Classification aims to assign an object to one of a set of predefined categories.
- Examples: image classification (cats, dogs), text classification (spam, non-spam).

- **Common techniques:**

- Convolutional Neural Networks (CNNs) for image classification.
- Recurrent Neural Networks (RNNs) and Transformers for text classification.

2. Regression:

- **Description:**

- Regression predicts a continuous value.
- Examples: predicting house prices, predicting temperature.

- **Common techniques:**

- Feedforward Neural Networks.

3. Object Detection:

- **Description:**

- Object detection not only classifies objects but also locates them within an image or video.
- Examples: detecting pedestrians in surveillance video, detecting traffic signs.

- **Common techniques:**

- R-CNN (Region-based Convolutional Neural Networks), YOLO (You Only Look Once), SSD (Single Shot MultiBox Detector).

The main problems addressed by deep learning

4. Image Segmentation:

- **Description:**
 - Image segmentation divides an image into meaningful regions, with each pixel assigned a label.
 - Examples: medical image segmentation to identify organs, satellite image segmentation for land analysis.
- **Common techniques:**
 - U-Net, FCN (Fully Convolutional Networks).

5. Natural Language Processing (NLP):

- **Description:**
 - NLP encompasses problems related to understanding and generating human language.
 - Examples: machine translation, sentiment analysis, text summarization, text generation.
- **Common techniques:**
 - RNNs, LSTM (Long Short-Term Memory), Transformers (BERT, GPT).

6. Generative Models:

- **Description:**
 - Generative models learn to create new data similar to the training data.
 - Examples: generating realistic images, generating music, generating text.
- **Common techniques:**
 - GAN (Generative Adversarial Networks), VAE (Variational Autoencoders).

Challenges in Deep Learning

1. Data Dependence:

- Requires massive, high-quality datasets, which are often costly and difficult to obtain.
- Data biases lead to biased models.

2. Computational Demands:

- Training deep networks is computationally intensive, needing powerful hardware.
- Long training times.

3. Model Complexity:

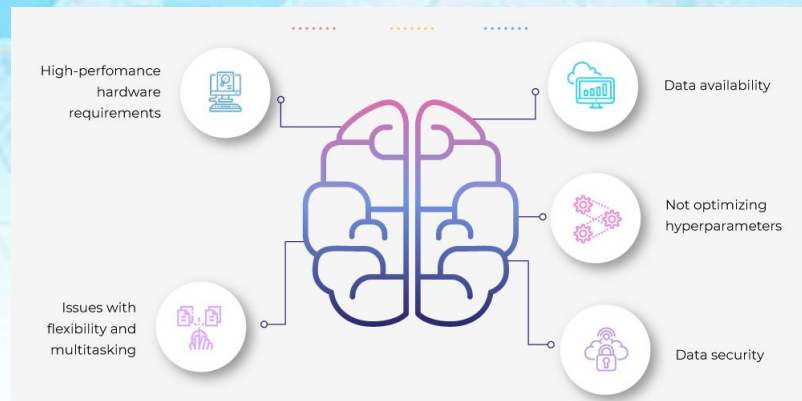
- "Black box" nature: lack of interpretability.
- Difficult hyperparameter tuning.
- Overfitting and underfitting issues.

4. Ethical Concerns:

- Potential for bias and unfairness.
- Privacy risks.
- Security risks from adversarial attacks.

5. Deployment difficulties:

- Hardware limitations on edge devices.
- Scalability problems.



Where to Find Deep Learning Exercises

- **Google Developers Machine Learning Crash Course:**
 - This course provides interactive exercises that allow you to experiment with neural network configurations.
 - It's a fantastic resource for beginners.
 - Here is the link: [Neural networks: Interactive exercises | Machine Learning - Google for Developers](#)
- **Kaggle:**
 - Kaggle is a platform that hosts data science competitions and provides datasets and notebooks.
 - You can find numerous deep learning exercises and tutorials on Kaggle.
 - Here is a good starting point: [Exercise: Deep Neural Networks - Kaggle](#)
- **Deep Learning Book Exercises:**
 - The book "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville has associated exercise material.
 - Here is a link to that resource: [Exercises - Deep Learning](#)
- **GitHub:**
 - GitHub is a treasure trove of deep learning projects and exercises.
 - Search for repositories related to deep learning tutorials or specific tasks.
 - Here is an example: [qiuyuanguong/Deep_Learning_Exercise: Deep Learning Exercise and Notebook - GitHub](#)

Dataset

Dataset is a collection of data. This data can come in various forms, such as: Numbers, Text, Images, Audio, Video. The key purpose of a dataset is to provide the raw material that machine learning algorithms use to learn patterns and make predictions.

Key Characteristics:

- **Structure.** Datasets can be structured (like tables in a database), semi-structured (like JSON or XML files), or unstructured (like raw text or images).
- **Size.** Datasets can range from small collections of a few data points to massive repositories containing billions of records.
- **Purpose.** Datasets are used for various purposes, including: Training machine learning models, testing model performance, analyzing data to gain insights
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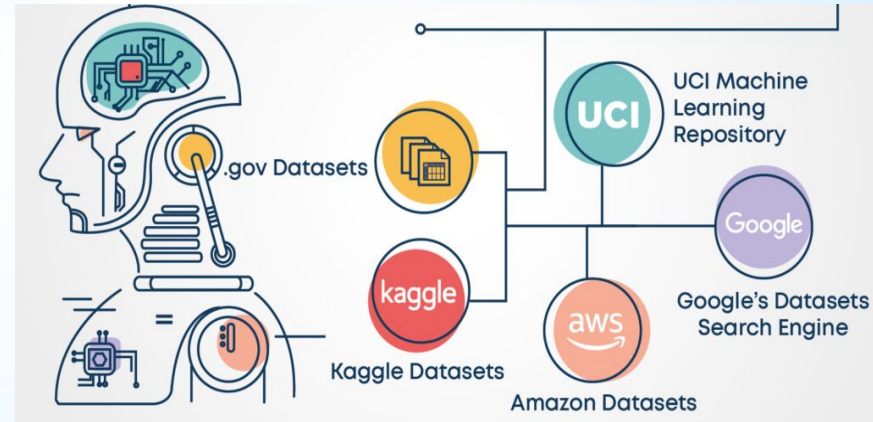
Importance in Deep Learning. Deep learning models, in particular, require large datasets to achieve high accuracy. The more data a model has, the better it can learn complex patterns.

Datasets are often divided into three subsets:

- **Training set:** Used to train the model.
- **Validation set:** Used to fine-tune the model's hyperparameters.
- **Testing set:** Used to evaluate the model's final performance.

Where to Find Datasets:

- **Kaggle:** A popular platform with a wide variety of datasets.
- **UCI Machine Learning Repository:** A classic source of datasets for machine learning research.
- **Google Dataset Search:** A search engine for datasets.
- **Hugging Face Datasets:** A library and hub for easily accessing and sharing datasets.
- **Government** open data portals (e.g., data.gov).



Dataset

Some common datasets perfect for beginners diving into deep learning

1. MNIST (Modified National Institute of Standards and Technology database):

- **Description:** A dataset of handwritten digits (0-9).
- **Use Case:** Image classification.
- **Why it's good:**
 - Very simple and well-understood.
 - Excellent for learning basic CNNs.
 - Abundant tutorials and examples.
- **Where to find it:** Built into many deep learning libraries (TensorFlow, PyTorch).

2. Fashion-MNIST:

- **Description:** Similar to MNIST, but with images of fashion items (e.g., shoes, shirts).
- **Use Case:** Image classification.
- **Why it's good:**
 - Slightly more challenging than MNIST.
 - Still relatively simple and easy to work with.
 - Replaces the original MNIST dataset for a more complex use case.
- **Where to find it:** Built into many deep learning libraries (TensorFlow, PyTorch).



Dataset

3. CIFAR-10 (Canadian Institute For Advanced Research):

- **Description:** A dataset of 60,000 32x32 color images in 10 classes (e.g., airplane, dog, cat).
- **Use Case:** Image classification.
- **Why it's good:**
 - A step up in complexity from MNIST.
 - Color images introduce more challenges.
 - Very popular.
- **Where to find it:** Built into many deep learning libraries (TensorFlow, PyTorch).

4. IMDB Movie Reviews:

- **Description:** A dataset of movie reviews, labeled as positive or negative.
- **Use Case:** Sentiment analysis (text classification).
- **Why it's good:**
 - Introduces natural language processing (NLP).
 - Good for learning RNNs and basic text processing.
 - Commonly used.
- **Where to find it:** Built into many deep learning libraries (TensorFlow, Keras).

5. Boston Housing Dataset:

- **Description:** Data about housing values in the Boston area, including features like crime rate and number of rooms.
- **Use Case:** Regression (predicting house prices).
- **Why it's good:**
 - Simple numerical dataset.
 - Good for learning basic regression models.
 - Easy to understand.
- **Where to find it:** Built into scikit-learn.

Reference

<https://atcold.github.io/NYU-DLSP20/>

https://www.youtube.com/watch?v=0bMe_vCZo30&list=PL80I41oVxglKcAHllsU0txr3OuTTaWX2v